

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

All core functional requirements have been successfully implemented and tested. The system provides accurate crop recommendations with a user-friendly interface and supports future enhancements such as weather integration, fertilizer recommendations, and crop yield prediction.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	AI-Powered Crop Recommendation	Predicts the most suitable crop based on soil nutrients and environmental conditions using Machine Learning.	Prediction Engine	Done	High
2	Data Preprocessing Pipeline	Cleans data, handles missing values, scales features, and prepares the dataset for model training.	Preprocessing Module	Done	Medium
3	Exploratory Data Analysis (EDA)	Generates visualizations and statistical insights to understand crop and soil patterns.	Data Analysis Module	Done	Medium
4	Machine Learning Model Training	Trains and evaluates Random Forest, Logistic Regression, and KNN models for crop prediction.	Model Development Module	Done	High
5	Confidence Score Generation	Displays prediction confidence percentages to improve decisionmaking reliability.	Prediction Engine	Done	Medium
6	Top Crop Matches	Provides alternative crop recommendations along with the best prediction.	Recommendation Module	Done	Medium

7	Responsive Web Interface	Enables users to enter soil and environmental parameters through an intuitive web application.	Frontend Module	Done	High
8	AI Assistant	Answers user queries regarding crop recommendations, dataset information, project features, and technical details.	AI Assistant Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Crop Recommendation Form, Results Dashboard
2	Backend / Logic	3	Crop Prediction Engine, Confidence Score Calculation, Error Handling
3	Database / Storage	1	Model and Scaler Storage using Pickle Files
4	API / Integration	1	AI Assistant Integration and Flask Route Handling

5	Security / Authentication	1	Input Validation and Secure Data Processing
---	---------------------------	---	---